

Introduction to the iPOP-UP HPC cluster

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Table of Contents

- 1 Introduction
- 2 The iPOP-UP@RPBS HPC cluster
- 3 In practice
- 4 Slurm usage
- 5 Modules
- 6 Conclusion

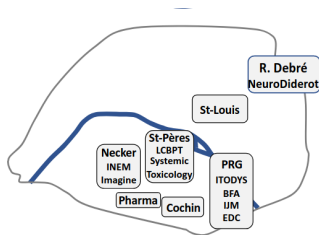


Who is this training for

- You work at Université Paris Cité
- You need (or might need) more computational power than you currently have
- You are familiar with Unix systems and Bash



The iPOP-UP project - <https://ipop.u-paris.fr/>



- iPOP-UP: Integrative Platform for Omics Projects at Université de Paris
- Multisite Bioinformatics platform
- Ranging from multiple 'Omics' techniques to structural and chemo- *in silico* Bioinformatics
- Compute nodes

The RPBS platform

<https://bioserv.rpbs.univ-paris-diderot.fr>



Missions:

- Development of Structural Bioinformatics methods
- Services deployment on the computing resource
- Expertise and training
- Hardware hosting

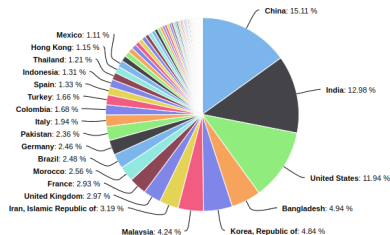
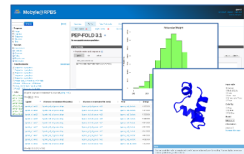
The RPBS platform

<https://mobile2.rpbs.univ-paris-diderot.fr:>

- free access
- 30+ Structural Bioinformatics services
- 1 million CPU hours /year
- 100,000 jobs /year

Other services:

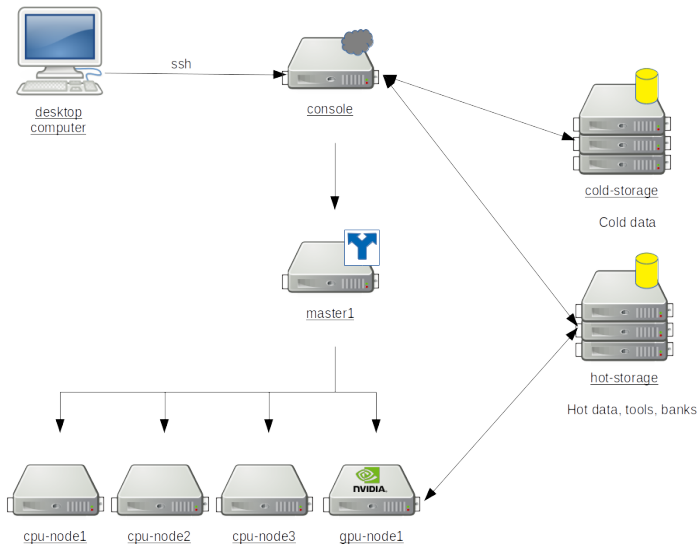
- SeamDock
- PEP-EDIT
- IA software: Alphafold 3, AlphaPullDown, Boltz-2, RFdiffusion, ProteinMPNN etc.



What is a HPC cluster for ?

- High hardware resources needs
- Long running analyses
- A lot of similar analyses
- Shared work between users

What is a HPC cluster ?



Computational hardware

Partitions (groups of compute nodes):

- ipop-up: 19 nodes, 1120 cores, 6 GPUs (A100)
- rpbs: 14 nodes, 472 cores, 8 GPUs (RTX2080Ti, A4500, 5000 Ada etc.)
- cmpli: 6 nodes, 156 cores, 14 GPUs (A100, RTX2080Ti, 6000, A6000, etc.)
- epigen: 4 nodes, 64 cores
- master-bi: 4 nodes, 112 cores, 15 GPUs (RTX6000, A30, L4)
- master-isdd: 2 nodes, 64 cores, 10 GPUs (A30, L4)

Storage:

- hot-storage: 125TB, very fast
- cold-storage: 240TB, very slow + backup

Virtualization servers, etc...



Connect using SSH (terminal)

Set your default project with:

```
[rey@ipop-up ~]$ set_project myproject
```

Note about ssh and security:

Your IP will be banned after 5 failed authentication attempts. Each public key count as one authentication attempt.

To disable public key authentication:

```
$ ssh -o PubkeyAuthentication=no rey@ipop-up.rpbs.univ-paris-diderot.fr
```

Connect using Open OnDemand (browser)

You can also access the cluster using the web interface Open OnDemand.

Open a web browser and go to

<https://ondemand.rpbs.univ-paris-diderot.fr>

Type in your login and password and log in.



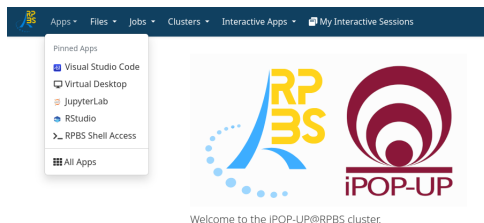
Log in to iPOP-UP@RPBS Cluster

Username

Password

Log in

Open OnDemand



Everything you need

- Browse and edit your files
- Launch your jobs
- Follow your jobs
- Use interactive apps

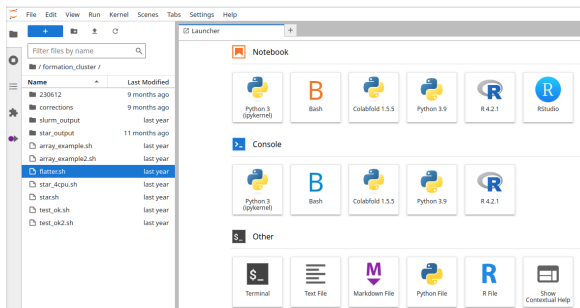
Available applications:

- Terminal
- Virtual Desktop (graphical tools - IGV)
- JupyterLab
- RStudio
- Visual Studio Code



Connect using JupyterHub (browser)

You can also access the cluster using the web interface JupyterHub (may be removed in the future). Open a web browser and go to <https://jupyterhub.rpbs.univ-paris-diderot.fr>
Select your project, the resources you need, and press Start.



Where you can go, write, or execute

User environments (very small, configuration files only)

/shared/home/username

Computations (hot data users conda environments)

/shared/projects/projectname

Processed data (cold data - on demand)

/cold-storage/username

Data banks (read-only)

/shared/banks/

Note about quotas

We operate quotas to limit the amount of disk space a group or a project can use on the hot storage filesystem. To check available space for all your projects:

```
[rey@ipop-up ~]$ statusBars
alphafold [###-----]      847 /   5000 GB
training  [-----]         0 /   1024 GB
```

```
[rey@ipop-up ~]$ lfsquotas training
Disk quotas for grp 6011 (gid 6011):
```

Filesystem	used	quota	limit	grace	files	quota	limit	grace
/shared/projects/training	96.26M	1T	2T	-	4	0	0	-

```
gid 6011 is using default file quota setting
```

Jobs **must** be launched from a project directory.

About Slurm



Slurm is the job scheduling system.

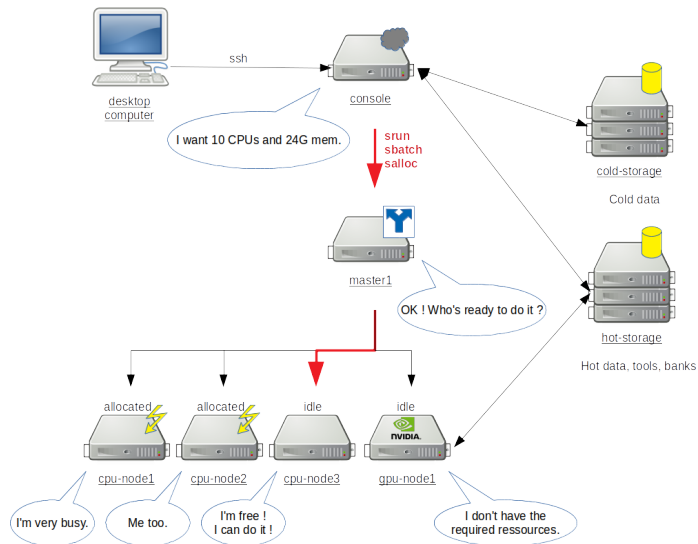
It is what will take your code and distribute it on the computing nodes, while ensuring they have the CPU(s) and RAM that you asked for.

It requires specific commands to run (srun, sbatch, salloc, etc...).

You **must** launch jobs with Slurm.



Flowsheet



srun

Launch a (simple) interactive job.

```
[rey@ipop-up ~]$ srun hostname
```

Some parameters can be added to the command line:

- partition**/–**p**: request a specific partition
- account**/–**A**: select the (project) account
- cpus-per-task**/–**c**: request that ncpus be allocated (default: 1 cpu)
- mem-per-cpu**: specify the required memory per cpu (default: 2GB)
- pty**: pseudo terminal mode

Example:

```
[rey@ipop-up ~]$ srun -A training -p ipop-up -c 8 hostname
```



sbatch

Launch more complex jobs.

myscript.sbatch

```
#!/bin/bash
#SBATCH --partition=ipop-up
#SBATCH --account=training
#SBATCH --cpus-per-task=8
#SBATCH --mem-per-cpu=4GB
#SBATCH --output=resultat.log
hostname
```

Example:

```
[rey@ipop-up ~]$ sbatch myscript.sbatch
```

squeue

List submitted jobs on the cluster:

```
[rey@ipop-up ~]$ squeue
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	ODELIST (REASON)
1008002	cmpli	argtoser	domingue	R	20:58:09	1	gpu-node14
993325	cmpli	calcam-p	ghoula	R	6-20:23:17	1	gpu-node8
993574	ipop-up	amyloid_	meuret	PD	0:00	1	(AssocGrpCPUMinutesLimit)
943019_[165-299]	ipop-up	kappa_va	badaoui	PD	0:00	1	(JobArrayTaskLimit)
943019_117	ipop-up	kappa_va	badaoui	R	7-17:55:27	1	cpu-node132

Some parameters can be added to filter jobs or show more infos:

- partition/-p**: specify the partition to view
- user/-u**: request jobs from a list of users
- format/-o**: specify the information to be displayed

Example:

```
[rey@ipop-up ~]$ squeue --me -p ipop-up
[rey@ipop-up ~]$ squeue -o "%10i %.9P %.8j %.12u %.15T %.15M %.15l %.6D %.20R %.6C %.10b %.10Q"
```

scancel

Kill a job

```
[rey@ipop-up ~]$ scancel job_id
```

Cancel all my jobs

```
[rey@ipop-up ~]$ scancel --me
```

sacct

Display accounting data for your running and completed jobs:

```
[rey@ipop-up ~]$ sacct --format=JobID,JobName,Start,Elapsed,NCPUS,NodeList,ReqMem,State --start 2023-06-01
→ --end 2023-06-30
```

JobID	JobName	Start	Elapsed	CPUTime	NCPUS	NodeList	ReqMem	State
1010778	sleep	2023-06-08T15:37:19	00:00:30	00:00:30	1	cpu-node133	2000Mc	COMPLETED

reportseff

Report a job efficiency

```
hennion@ipop-up]$ module load reportseff
hennion@ipop-up]$ reportseff 40913604
```

JobID	State	Elapsed	TimeEff	CPUEff	MemEff
40913604	CANCELLED	00:00:37	0.0%	16.2%	20.6%

Analyse a folder of slurm outputs

```
hennion@ipop-up]$ reportseff --format "+Start,CPUTime,NCPUS,NodeList,MaxRSS,ReqMem" --modified-sort
→ slurm_output/
```

JobID	State	Elapsed	TimeEff	CPUEff	MemEff	Start	CPUTime	NCPUS	NodeList	MaxRSS	ReqMem
multiqc--41907184.out	COMPLETED	00:00:07	2.3%	57.1%	1.0%	2024-09-24T12:23:17	00:00:07	1	cpu-node-45	3184K	191M
fastqc-sample-SRR3099585_chr18-41907179.out	COMPLETED	00:00:25	8.3%	64.0%	1.1%	2024-09-24T12:22:47	00:00:25	1	cpu-node-38	3284K	286M
fastqc-sample-SRR3105697_chr18-41907180.out	COMPLETED	00:00:15	5.0%	80.0%	1.1%	2024-09-24T12:22:47	00:00:15	1	cpu-node-39	3268K	286M
fastqc-sample-SRR3099587_chr18-41907181.out	COMPLETED	00:00:14	4.7%	85.7%	1.1%	2024-09-24T12:22:47	00:00:14	1	cpu-node-40	3208K	286M
fastqc-sample-SRR3105698_chr18-41907183.out	COMPLETED	00:00:13	4.3%	69.2%	1.1%	2024-09-24T12:22:47	00:00:13	1	cpu-node-48	3192K	286M
fastqc-sample-SRR3099586_chr18-41907178.out	COMPLETED	00:00:12	4.0%	66.7%	1.1%	2024-09-24T12:22:47	00:00:12	1	cpu-node-45	3208K	286M
fastqc-sample-SRR3105699_chr18-41907182.out	COMPLETED	00:00:12	4.0%	66.7%	1.1%	2024-09-24T12:22:47	00:00:12	1	cpu-node-48	3268K	286M
multiqc--41907169.out	OUT_OF_MEMORY	00:00:06	2.0%	16.7%	3.3%	2024-09-24T12:21:06	00:00:06	1	cpu-node-45	3240K	95M
fastqc-sample-SRR3099585_chr18-41907159.out	COMPLETED	00:00:33	11.0%	54.5%	75.7%	2024-09-24T12:20:25	00:00:33	1	cpu-node-38	221708K	286M
fastqc-sample-SRR3105697_chr18-41907160.out	COMPLETED	00:00:19	6.3%	73.7%	1.1%	2024-09-24T12:20:25	00:00:19	1	cpu-node-39	3212K	286M
fastqc-sample-SRR3105698_chr18-41907163.out	COMPLETED	00:00:17	5.7%	82.0%	1.1%	2024-09-24T12:20:25	00:00:17	1	cpu-node-40	3224K	286M
fastqc-sample-SRR3099587_chr18-41907161.out	COMPLETED	00:00:15	5.0%	80.0%	1.1%	2024-09-24T12:20:25	00:00:15	1	cpu-node-40	3192K	286M
fastqc-sample-SRR3105699_chr18-41907162.out	COMPLETED	00:00:14	4.7%	71.4%	1.1%	2024-09-24T12:20:25	00:00:14	1	cpu-node-48	3244K	286M
fastqc-sample-SRR3099586_chr18-41907158.out	COMPLETED	00:00:15	5.0%	53.3%	1.1%	2024-09-24T12:20:25	00:00:15	1	cpu-node-45	3304K	286M
multiqc--41907141.out	COMPLETED	00:00:24	0.6%	25.0%	0.6%	2024-09-24T12:15:57	00:00:24	1	cpu-node-38	3228K	580M
fastqc-sample-SRR3105698_chr18-41907138.out	COMPLETED	00:00:18	0.2%	72.2%	0.6%	2024-09-24T12:15:35	00:00:18	1	cpu-node-34	3244K	580M
fastqc-sample-SRR3105699_chr18-41907137.out	COMPLETED	00:00:16	0.2%	68.8%	0.6%	2024-09-24T12:15:35	00:00:16	1	cpu-node-34	3228K	580M
fastqc-sample-SRR3099585_chr18-41907133.out	COMPLETED	00:00:15	0.2%	66.7%	0.6%	2024-09-24T12:15:35	00:00:15	1	cpu-node-39	3144K	580M
fastqc-sample-SRR3099585_chr18-41907134.out	COMPLETED	00:00:15	0.2%	73.3%	0.6%	2024-09-24T12:15:35	00:00:15	1	cpu-node-40	3272K	580M
fastqc-sample-SRR3099587_chr18-41907136.out	COMPLETED	00:00:14	0.2%	71.4%	0.6%	2024-09-24T12:15:35	00:00:14	1	cpu-node-48	3208K	580M
fastqc-sample-SRR3105697_chr18-41907135.out	COMPLETED	00:00:15	0.2%	66.7%	0.6%	2024-09-24T12:15:35	00:00:15	1	cpu-node-40	3232K	580M

Job arrays

Job arrays offer a mechanism for launching a lot of tasks at the same time. Each task of the job will have the environment variable `$SLURM_ARRAY_TASK_ID` set to its array index value.

`myscript.sbatch`

```
#!/bin/bash
#SBATCH --partition=ipop-up
#SBATCH --account=training
#SBATCH --output=resultat_%a.log
#SBATCH --array=1-3
case "$SLURM_ARRAY_TASK_ID" in
  1) fruit='orange';;
  2) fruit='apple';;
  3) fruit='banana';;
esac
echo $fruit
```

Job arrays

```
[rey@ipop-up ~]$ sbatch myscript.sbatch
```

Results:

```
[rey@ipop-up ~]$ ls
resultat_1.log  resultat_2.log  resultat_3.log
[rey@ipop-up ~]$ tail resultat_*
==> resultat_1.log <==
orange

==> resultat_2.log <==
apple

==> resultat_3.log <==
banana
```

Job arrays

Another example:

myscript.sbatch

```
#!/bin/bash
#SBATCH --partition=ipop-up
#SBATCH --account=training
#SBATCH --output=resultat_%a.log
#SBATCH --array=0-2000%50
#SBATCH --cpus-per-task=2
LINE=$(sed -n ${SLURM_ARRAY_TASK_ID}p list.txt)
fastqc $LINE
```

Job arrays

```
#SBATCH --array=0-15
```

= 16 jobs (\$SLURM_ARRAY_TASK_ID: from 0 to 15 included).

```
#SBATCH --array=10-16:2
```

= 4 jobs (\$SLURM_ARRAY_TASK_ID: 10,12,14,16).

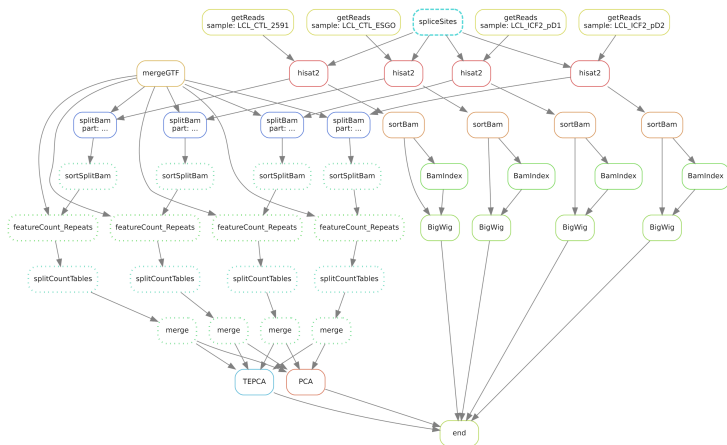
```
#SBATCH --array=2,3-7:2,11,13
```

= 6 jobs (\$SLURM_ARRAY_TASK_ID: 2,3,5,7,11,13).

```
#SBATCH --array=1-10000%32
```

= 10 000 jobs, max 32 simultaneous jobs

Complex workflows



Use workflow managers such as Snakemake or Nextflow.

Modules



Tools are installed on the cluster in virtual environments

- each tool has its own dependencies (libraries) and it's not possible to make them all coexist in the same environment
- reproducibility: keep several versions of a tool



Conda environments



Containers (Apptainer)



Modules

They can be loaded with the module command.

Look for the different versions of multiqc:

```
[rey@ipop-up ~]$ module av multiqc  
multiqc/1.3 multiqc/1.6 multiqc/1.7 multiqc/1.9
```

Load an environment:

```
[rey@ipop-up ~]$ module load multiqc/1.9  
[rey@ipop-up ~]$ which multiqc  
/shared/software/conda/envs/multiqc-1.9/bin/multiqc
```

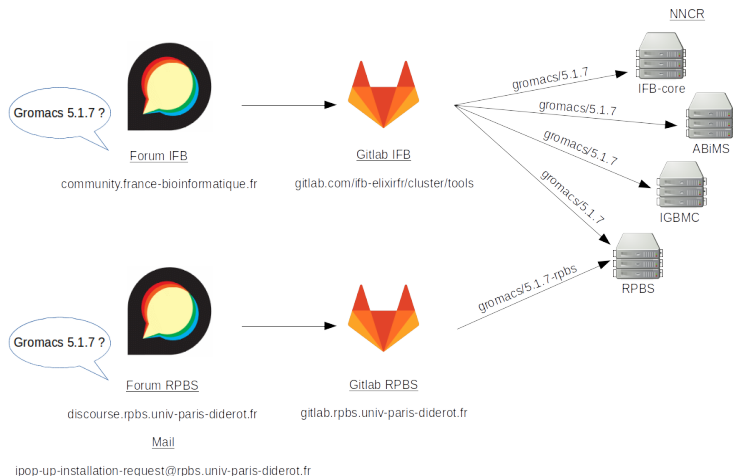
List loaded environments:

```
[rey@ipop-up ~]$ module list  
Currently Loaded Modulefiles:  
  1) multiqc/1.9      2) blast/2.13.0
```

Unload environments:

```
[rey@ipop-up ~]$ module purge
```

How tools are installed



Useful resources

To find out more, the SLURM manual : `man sbatch` or
<https://slurm.schedmd.com/sbatch.html>

Ask for help or signal problems on the cluster :
<https://discourse.rpbs.univ-paris-diderot.fr/>

iPOP-UP cluster documentation:
<https://ipop-up.docs.rpbs.univ-paris-diderot.fr/documentation/>

iPOP-UP cluster getting started tutorial (fr):
<https://ipop.u-paris.fr/documentation-du-cluster-ipop-uprpbs/>



Exercices

<https://tinyurl.com/ipop-up-2026>



https://parisepigenetics.github.io/bibs/training/cluster_training_2602